(d)	**Light and sound**
Students should:	
3.14	know that light waves are transverse waves and that they can be reflected and refracted
3.15	use the law of reflection (the angle of incidence equals the angle of reflection)
3.16	draw ray diagrams to illustrate reflection and refraction
3.17	*practical: investigate the refraction of light, using rectangular blocks, semi-circular blocks and triangular prisms*
3.18	know and use the relationship between refractive index, angle of incidence and angle of refraction: $n = \frac{\sin i}{\sin r}$
3.19	*practical: investigate the refractive index of glass, using a glass block*

Students should:	
3.22	know and use the relationship between critical angle and refractive index: $\sin c = \frac{1}{n}$
3.23	know that sound waves are longitudinal waves that can be reflected and refracted
3.24P	**know that the frequency range for human hearing is 20–20 000 Hz**
3.25P	***practical: investigate the speed of sound in air***
3.26P	**understand how an oscilloscope and microphone can be used to display a sound wave**
3.27P	***practical: investigate the frequency of a sound wave using an oscilloscope***
3.28P	**understand how the pitch of a sound relates to the frequency of vibration of the source**
3.29P	**understand how the loudness of a sound relates to the amplitude of vibration of the source**